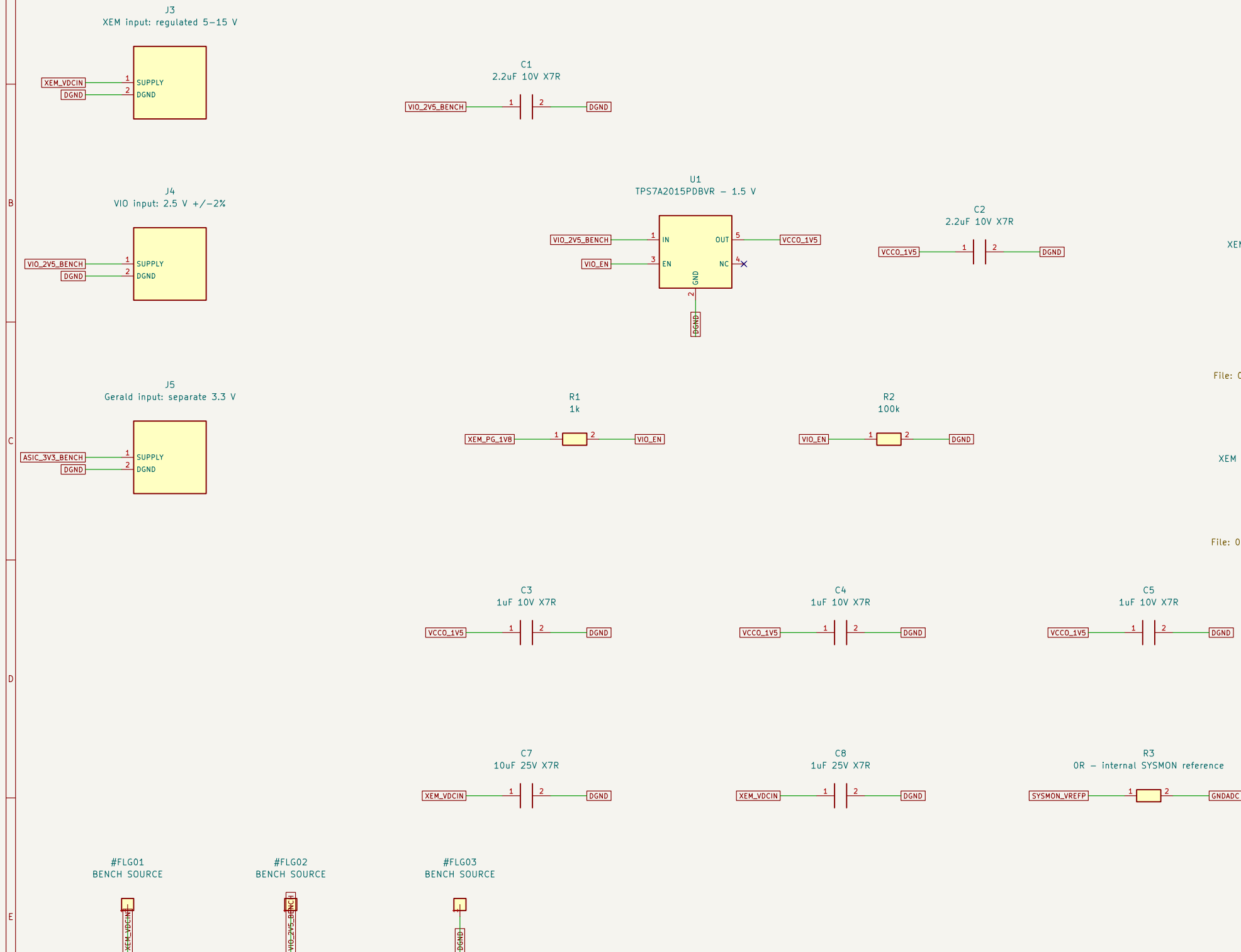


Confirmed plan: reuse Gerald LDO + existing ASIC; rigid adapter directly to XEM8305.

This board routes data and generates FPGA VCCO. Gerald board retains ASIC regulators.

No micro-HDMI interface is required for this backup path.



LDO J1 to XEM MC1: 23 signals

File: 01_LDO_And_MC1.kicad_sch

XEM MC2 power and LDO J19 support

File: 02_MC2_And_Mechanical_J19.kicad_sch

XEM MC3: fitted socket and reserved GTH

File: 03_MC3_Reserved_High_Speed.kicad_sch

C3/C4/C5/C6: place at bank64/84/85/86 power contacts; preserve effective $\geq 1\mu\text{F}$.

Bring-up order: ASIC and stimulation supplies OFF -> XEM -> stable 1.5V VCCO -> safe GPIO -> ASIC.

Power down ASIC before XEM. Direct paths provide NO automatic unpowered–signal isolation.

Existing high-voltage supply and AGND arrangement must be reviewed with Gerald; not invented here.

VIO planning envelope: 1–150 mA aggregate at ≤ 55 C ambient; physical load/thermal proof open.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /
File: LDO_to_XEM8305_Direct_R2.kicad_sch

Title: LDO to XEM8305: bench power

Size: A3

Date: 2026-10-05

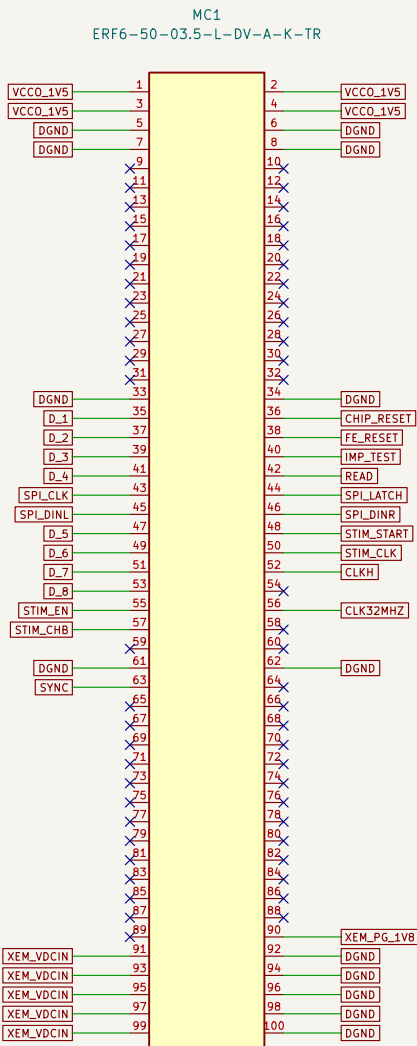
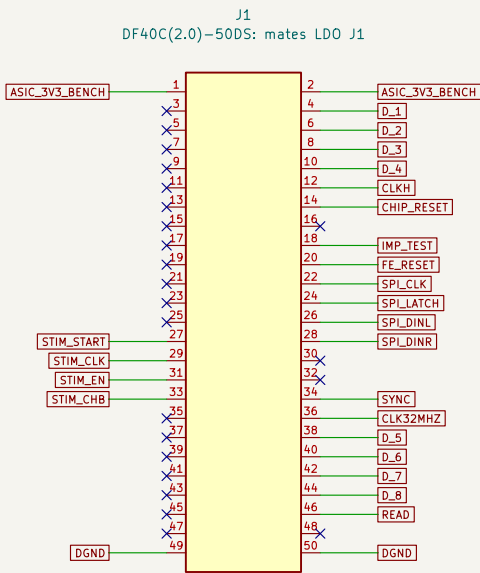
Rev: R2 candidate

Id: 1/4

Howard confirmed existing J1 + J19 on 2 October; J2 remains ASIC-facing.

23 signal routes preserved. FPGA pin/bank assignment is a proposal verified against official CSV.

Digital 1.5 V limits, direction, timing and reset/stim inactive levels still require confirmation.



Existing J1 pins47/48 have only an isolated VCC pair: adapter intentionally NC.

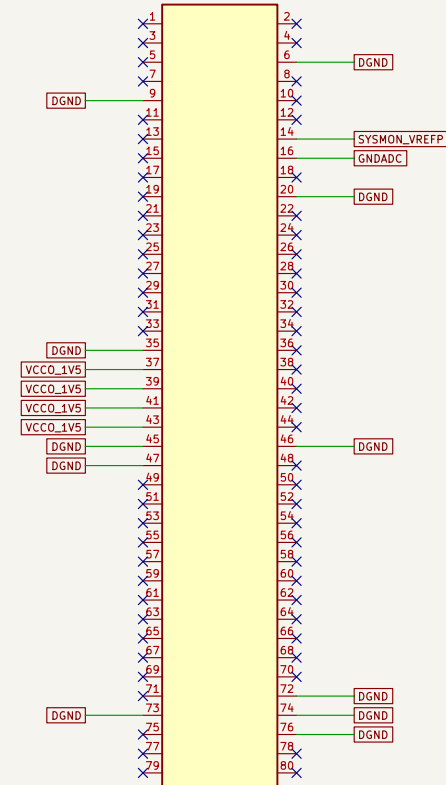
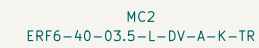
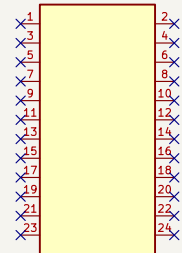
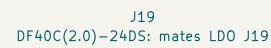
All assigned signal balls use Bank64. CLKH/SYNC/CLK32MHZ use clock-capable sites.

No claim that source4cmPCB experience validates this longer route or eight-chip system.

No GTP/GTH or level translation is used for these nominal1.5VGPIO paths.

J19 is a confirmed mating connector; every contact remains electrically unused.

Do not assign power to J19 without a revised Gerald PCB pin map.



All three expansion sockets are fitted. MC3 high-speed signals are explicitly reserved.

All four external VCCO banks share the proposed 1.5V rail. Verify actual bank currents.

Module1.8V is an output, not a3.3V ASIC source. No tie to bench3.3V or other outputs.

The adapter is a separate board. Do not edit or rewire the supplied LDO project.

Bench candidate: startup / timing / mechanics unverified

FPGA PCB project

Sheet: /XEM MC2 power and LDO J19 support/

File: 02_MC2_And_Mechanical_J19.kicad_sch

Title: XEM MC2 power and LDO J19 support

Size: A3

Date: 2026-10-05

Rev: R2 candidate

KiCad E.D.A. 10.0.6

Id: 3/4

